

Abbreviations and Symbols

A	ampere
Å	angstrom
ABC	activity-based costing
ac	alternating current
ACGIH	American Conference of Governmental Industrial Hygienists
AES	Auger electron spectroscopy
AMS	Aerospace Material Specification
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
at. %	atomic percent
atm	atmosphere (pressure)
bal	balance
bcc	body-centered cubic
BEI	back scattering electron image
bet	body-centered tetragonal
Btu	British thermal unit
CIP	cold isostatic pressing
cm	centimeter
CMC	ceramic-matrix composite
CNC	computer numerical control
CP	commercial pure
cpm	cycles per minute
cps	cycles per second
CSA	Canadian Standards Association
cSt	centiStokes
CTE	coefficient of thermal expansion
CVD	chemical vapor deposition
CVI	chemical vapor infiltration (impregnation)
d_a	project area diameter of the particle
dB	decibel
Dc	direct current
DC arc	direct current arc
DCA	differential calorimetric analysis
diam	diameter
DIN	Deutsche Institut fur Normung
d_p	perimeter diameter of the particle
DTA	differential thermal analysis
e	charge of an electron; natural log base, 2.71828
E	modulus of elasticity; Young's modulus; potential
E_s	secant modulus
EDM	electrical discharge machining
EIMAC	European Industry Metals and Alloys Classification Group
El	elongation
emf	electromotive force
EMI	electromagnetic interference
Eq	equation
EPMA	European Powder Metallurgy Association

ESC	environmental stress cracking
esu	electrostatic units
et al.	and others
eV	electron volt
F	Faraday constant; force
fcc	face-centered cubic
FCGR	fatigue crack growth rate
fct	face-centered tetragonal
Fig.	figure
FM	figure of merit
ft	foot
g	gram
G	gauss
GA	gas atomized
gal	gallon
GPa	gigapascal
gpd	grains per denier
gr	grain
h	hour
h	thickness
H	henry
H	change in height; degree of homogenization; enthalpy; hardness; height; magnetic field
HB	Brinell hardness
hcp	hexagonal close-packed
HDT	heat-deflection temperature
HDVC	hot dynamic non-capsule vacuum compacting
HE	hot extrusion
HERF	high-energy-rate forging
HIP	hot isostatic pressing
hp	horsepower
HPWA	high pressure water atomization
HR	Rockwell hardness
HV	Vickers hardness
Hz	hertz
i	current (measure of number of electrons)
I	current; emergent intensity
IACS	International annealed copper standard
IATA	International air transport association
IC	integrated circuit
ID	inside diameter
IEC	International Electrotechnical Commission
IMDG Code	International maritime dangerous goods code
in.	inch
INTER	International thermonuclear experimental reactor
IPD	integrated product development
IPMD	International Powder Metallurgy Directory
IPMS	Institute for Problems of Materials Science
IPPC	integrated pollution prevention and control
IPTS	International Practical Temperature Scale
IR	infrared (radiation)
ISAO	International Civil Aviation Organization
ISO	International Organization for Standardization
ITS	International Temperature Scale
IV	intrinsic viscosity
J	joule
JIS	Japanese Industrial Standard
JWRL	Welding Research Institute of Osaka University
k	karat

<i>k</i>	Boltzmann constant; notch sensitivity factor; thermal conductivity; wave number
K	Kelvin
<i>K</i>	bulk modulus of elasticity; coefficient of thermal conductivity; empirical constant; interface reaction; mean integrated thermal conductivity
<i>K_I</i>	stress-intensity factor
<i>K_{Ic}</i>	mode I critical stress-intensity factor; plane-strain fracture toughness
<i>K_{Id}</i>	dynamic fracture toughness
<i>K_{Isc}</i>	threshold stress intensity for stress-corrosion cracking
<i>K_c</i>	plane-stress fracture toughness
<i>K_f</i>	stress-concentration factor
kg	kilogram
km	kilometer
kN	kilonewton
kPa	kilopascal
ksi	kips (1000 lb) per square inch
kV	kilovolt
kW	kilowatt
<i>l</i>	mean free path; length
L	liter; longitudinal direction; liquid
lb.	pound
lbf	pound force
LCA	life cycle analysis (or assessment)
LCIL	low concentration ignition limit
<i>L/D</i>	length-to-diameter ratio
LED	light-emitting diode
LEFM	linear-elastic fracture mechanics
LOC	explosion proof limit oxygen content
ln	natural logarithm (base <i>e</i>)
log	common logarithm (base 10)
m	matrix; meter
<i>m</i>	ion mass; Weibull modulus
M	metal atom
<i>M</i>	molecular weight
MA	mechanical alloying
mA	milliampere
MeV	megaelectronvolt
NDT	nondestructive testing
mg	milligram
Mg	megagram
MIM	metal injection molding
min	minimum; minute
MJ	megajoule
mL	milliliter
MLCCs	multilayer ceramic capacitors
MILT	marketing lead time
mm	millimeter
MMC	metal-matrix composite
mod	modified
mol%	mole percent
MOR	modulus of resilience; modulus of rupture
mPa	millipascal
MPa	megapascal
mpg	miles per gallon
mph	miles per hour
MPIF	Metal Powder Industrial Federation
ms	millisecond
MS	megasiemens
MSDS	material safety data sheets

mT	millitesla
MTBT	mean time between failures
MTTF	mean time to failure
mV	millivolt
MV	megavolt
MW	molecular weight
n	growth exponent
<i>n</i>	integral number
N	Newton
<i>N</i>	fatigue life (number of cycles)
NAERG	North American Emergency Response Guidebook
NASA	National Aeronautics and Space Administration
NBS	National Bureau of Standards
NC	numerical control
NDE	nondestructive evaluation
NDI	nondestructive inspection
mg	milligram
Mg	megagram
MIM	metal injection molding
NFPA	National Fire Protection Association
NIOSH	National Institute for Occupational Safety and Health
NIST	National Institute of Standards and Technology
NJDHSS	New Jersey Department of Health and Senior Services
nm	nanometer
No.	number
NSS	near-net shape
OD	outside diameter
ODS	oxide-dispersion strengthened
Oe	oersted
OELs	occupational exposure limits
OHA	operating hazards analysis
ORNL	Oak Ridge National Laboratory
OSHA	Occupational Safety and Health Administration
oz	ounce
p	page
<i>p</i>	pressure
Pa	pascal
PCA	process control agent
PCE	pyrometric cone equivalent
PDS	product design specification
PDT	product design team
PEL	permissible exposure limit
P/F	powder forging
pH	negative logarithm of hydrogen-ion activity
PM	powder metallurgy
ppb	parts per billion
ppba	parts per billion atomic
ppm	parts per million
ppt	parts per trillion
Pr	Prandtl number
psi	pounds per square inch
PTC	positive temperature coefficient
PVD	physical vapor deposition
QC	quality control; quasicrystals
QLD	quality loss function
r	radius vector in a plane normal to the axis
<i>r</i>	particle radius; radius of curvature; rate of reaction; reflectivity; spherical particle radius
R	roentgen

R	average particle radius; gas constant; radius; ratio of the minimum stress to the maximum stress; resistance; reliability
R_a	surface roughness in terms of arithmetic average
RA	reduction of area rad absorbed radiation
rad	absorbed radiation dose; radian
Re	Reynolds number
REM	rare earth metal
Ref	reference
rem	remainder
rf, RF	radio frequency
RH	relative humidity
rms, RMS	root mean square
rpm	revolutions per minute
RS	rapid solidification
RT	room temperature
RTI	relative thermal index
s	second
S	siemens
S	distance traveled
SAE	Society of Automotive Engineers
scfm	standard cubic foot per minute
SCOEL	Scientific Committee on Occupational Exposure Limits
SEM	scanning electron microscopy
sfm	surface feet per minute
SG	standard grade
SI	Système International d'Unités
sineh	sine hyperbolic
S/N	signal-to-noise (ratio)
S-N	stress-number of cycles
SPC	statistical process control
sp gr	specific gravity
SPF	superplastic forming
std	standard
STEM	scanning transmission electron microscopy
STM	scanning tunneling microscopy
t	thickness; time
T	Tesla; transverse direction
T	absolute temperature; temperature; tenacity; total dispersion; transmittance
T_m	melt temperature
TEM	transmission electron microscopy
TGA	thermogravimetric analysis
TLVs	threshold limit values
TM	threshold limit values; transition metal
TMA	thermomechanical analysis
TMS	The Minerals, Metals, and Materials Society
TP	thermoplastic
TQM	total quality management
TWA	time-weighted average
USBM	US Bureau of Mines
NJDHSS	US New Jersey Department of Health and Senior Services
UTS	ultimate tensile strength
UV	ultraviolet
v	velocity; workpiece
V	volt
VAMI	Russian National Aluminum-Magnesium Institute
VD	vacuum degassing
VHP	vacuum hot pressing
V_f	volume fraction of fiber
V_m	volume fraction of matrix

V_v	volume fraction of void content
VI	viscosity index
vol	volume
vol%	volume percent
W	watt
W	total radiation; width
WA	water atomization, water atomized
WELs	workplace exposure limits
wt%	weight percent
x	axial distance
X	neck diameter
XRD	X-Ray diffraction
Y	scale of microstructural segregation
yr	year
YS	yield strength
z	ion charge
Z	atomic number; impedance
θ	angular measure; degree
$^{\circ}\text{C}$	degree Celsius (centigrade)
$^{\circ}\text{F}$	degree Fahrenheit
0^0	fiber direction
90^0	perpendicular to fiber direction
α	coefficient of thermal expansion
γ	shear strain; surface energy; surface tension
γ_b	interfacial energy
δ	an increment; a range; change in quantity; grain boundary width; median diameter
Δ	an increment; a range; change in quantity
ΔT	temperature difference or change
ε	strain
$\dot{\varepsilon}$	strain rate
ζ	local resistance to flow of gas (air)
η	loss coefficient; viscosity
θ	angle; geometrical constant
θ_i	angle of incidence
θ_r	angle of refraction
κ	dielectric constant
λ	thermal conductivity
μ	friction coefficient; linear attenuation coefficient; magnetic permeability
μin	microinch
μm	micrometer (micron)
μs	microsecond
ν	Poisson's ratio; velocity
π	pi (3.141592)
ρ	density; resistivity
σ	Stefan-Boltzmann constant; strength; stress; tensile stress; tensile strength
σ_a	applied stress
σ_f	applied axial load
Σ	summation of
τ	shear stress
ϕ	angle of internal friction; dihedral angle; porosity; power
Φ	reaction rate (kinetic function); volume concentration of impurity ions
ψ	damping; geometrical constant
ω	frequency
Ω	atomic volume; electrical conductivity; molecular volume of solute; ohm
∂	partial derivative
$\partial\eta/\partial t$	kinetic function
$\leftrightarrow$	direction of reaction
$=$	equals

$\approx$	approximately equals
$\neq$	not equal to
$\equiv$	identical with
$>$	greater than
$>>$	much greater than
$\geq$	greater than or equal to
∞	infinity
$\propto$	is proportional to; varies as
$\int$	integral of
$<$	less than
$<<$	much less than
$\leq$	less than or equal to
$\pm$	maximum deviation
$-$	minus; negative ion charge
$\times$	diameters (magnification); multiplied by
$\cdot$	multiplied by
$/$	per
$\%$	percent
$+$	plus; positive ion charge
$\sqrt{}$	square root of
$\sim$	approximately; similar to

GREEK ALPHABET

A, α	alpha
B, β	beta
Γ, γ	gamma
Δ, δ	delta
E, ϵ	epsilon
Z, ζ	zeta
H, η	eta
Θ, θ	theta
I, ι	iota
K, κ	kappa
Λ, λ	lambda
M, μ	mu
N, ν	nu
Ξ, ξ	xi
Π, π	pi
P, ρ	rho
Σ, σ	sigma
T, τ	tau
Y, υ	upsilon
Φ, ϕ	phi
X, χ	chi
Ψ, ψ	psi
Ω, ω	omega